

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built an **AI orchestration workflow** for company-registration drafts, combining structured prompt outputs, operator validation, public-site API lookups, and automation handoff.
- Implemented async backend execution with retries, session recovery, browser automation, and state validation, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, covering preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output pipeline issues across conformation, label mapping, PyTorch/ONNX-style inference, and binary mask generation inside a mature open-source codebase.

Open Source

[Contributions](#) 

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Swish Support System | *JavaScript, React, Python, PostgreSQL/pgvector, Redis, Langfuse*

[GitHub](#) 

- Built a **hybrid AI + rule-based logic system** where LLM assessments handle messy complaint semantics while deterministic policy code controls refunds, replacements, and escalations.
- Implemented **RAG-style context grounding** over order, item, evidence, customer, and policy data before each traceable assessment and response-generation step.
- Added agent memory/state management, Langfuse traces, response-quality checks, and regression suites for multi-turn cases where context must survive clarifications and follow-ups.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, JSONL*

[GitHub](#) 

- Built an **AI orchestration system** with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and JSONL traces.
- Implemented **vector memory** and semantic lesson retrieval so agents could pull similar past decisions into context and inspect retrieval quality through validation harnesses.
- Added reward-loop style evaluation and checkpointed runs so agent behavior, retrieved context, and final decisions could be compared across iterations.

Seaweed: Contest Platform | *TypeScript, React, Next.js, Go, PostgreSQL, Judge0, Firebase*

[GitHub](#) 

- Built the **React/Next.js assessment UI** with auth-gated problem pages, code editor, submissions, admin review/shortlisting, and live ranked leaderboards.
- Implemented Go judging APIs with Judge0, PostgreSQL rankings/materialized views, and Redis-backed runtime state for **500+ concurrent users** and **sub-5s P95 verdict latency**.

Technical Skills

Languages: JavaScript, TypeScript, Python, Go

Frameworks: React, Next.js, Node.js, GraphQL, Vite, FastAPI, Gin, Echo, PyTorch

Tools: PostgreSQL/pgvector, Redis, SQLite, Vector Memory, Firebase Auth, Judge0, Docker, GCP, GitHub Actions, Langfuse, Prometheus